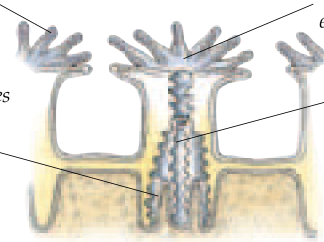
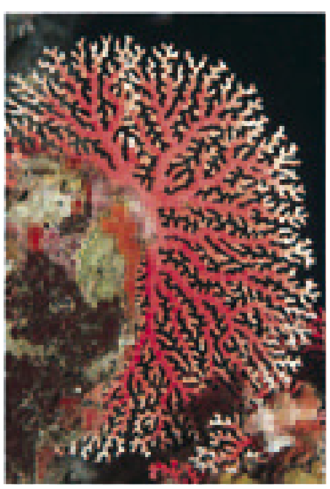
The coral kingdom

In the crystal clear, warm waters of the tropics, coral reefs flourish, covering vast areas. Made of the skeletons of stony corals, coral reefs are cemented together by chalky algae. Most stony corals are colonies of many tiny, anemonelike individuals, called polyps. Each polyp makes its own hard limestone cup (skeleton) which protects its soft body. To make their skeletons, the coral polyps need the help of microscopic, single-celled algae that live inside them. The algae need sunlight to grow, which is why coral reefs are found only in sunny, surface waters. In return for giving the algae a home, corals get some food from them but also capture plankton with their tentacles. Only the upper layer of a reef is made of living corals, which build upon skeletons of dead polyps. Coral reefs are also home to soft corals and sea fans, which do not have stony skeletons. Related to sea anemones and jellyfish, corals grow in an exquisite variety of shapes (mushroom, daisy, staghorn) and some have colorful skeletons.



INSIDE A CORAL ANIMAL
In a hard coral, a layer of tissue joins each polyps to its neighbor. To reproduce, they divide in two or release eggs and sperm into the water.



STINGING CORAL
Colorful hydrocorals are related to sea firs and, unlike horny and stony corals, produce jellyfishlike forms that carry their sex organs. Known as fire corals, they have potent stings on their polyps.

BLACK CORAL
In living black corals, the skeleton provides support for the living tissues and the branches bear rows of anemonelike polyps. Black corals are mainly found in tropical waters, growing in the deep part of coral reefs. Although they take a long time to grow, the black skeleton is sometimes used to make jewelry.